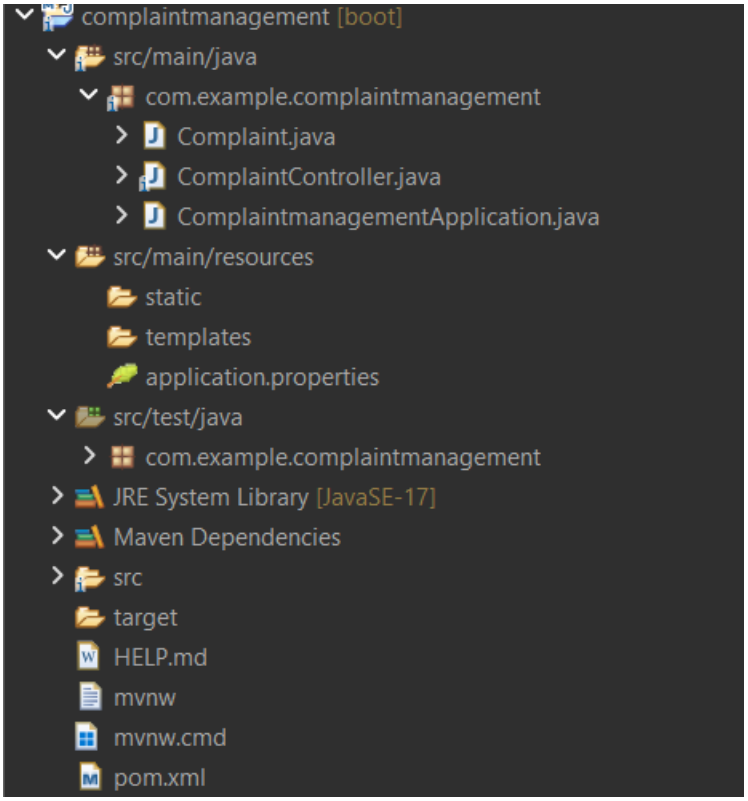
Java spring boot

Case study –5

Complaint Management System

1 project structure



2 Complaint.java

```java

package com.example.complaintmanagement;

public class Complaint {

    private Integer id;
    private String description;
    private String category;
    private String status;

    public Complaint() {
    }

    public Complaint(Integer id, String description, String category) {
        this.id = id;
        this.description = description;
        this.category = category;
        this.status = "PENDING";
    }

    public Integer getId() {
        return id;
    }

    public void setId(Integer id) {
        this.id = id;
    }

    public String getDescription() {
        return description;
    }

    public void setDescription(String description) {
        this.description = description;
    }

    public String getCategory() {
        return category;
    }

```

```java

    public void setCategory(String category) {
        this.category = category;
    }

    public String getStatus() {
        return status;
    }

    public void setStatus(String status) {
        this.status = status;
    }
}

```

3 ComplaintController.java

```java
package com.example.complaintmanagement;

import java.util.ArrayList;
import java.util.List;

import org.springframework.web.bind.annotation.*;

@RestController
public class ComplaintController {

    private List<Complaint> complaints = new ArrayList<>();
    private int nextId = 1;

    // Register a new complaint
    @PostMapping("/complaints")
    public Complaint addComplaint(@RequestBody Complaint complaint) {

        complaint.setId(nextId++);
        complaint.setStatus("PENDING");

        complaints.add(complaint);

        return complaint;
    }

    // Retrieve all complaints
    @GetMapping("/complaints")
    public List<Complaint> getAllComplaints() {
        return complaints;
    }

    // Update complaint status
    @PutMapping("/complaints/{id}")
    public Complaint updateComplaint(
            @PathVariable Integer id,
            @RequestBody Complaint updatedComplaint) {

        for (Complaint complaint : complaints) {
            if (complaint.getId().equals(id)) {
                complaint.setDescription(updatedComplaint.getDescription());
                complaint.setCategory(updatedComplaint.getCategory());
                complaint.setStatus(updatedComplaint.getStatus());

                return complaint;
            }
        }

        return null;
```

```java
    // Delete a complaint
    @DeleteMapping("/complaints/{id}")
    public String deleteComplaint(@PathVariable Integer id) {

        for (Complaint complaint : complaints) {
            if (complaint.getId().equals(id)) {
                complaints.remove(complaint);
                return "Complaint deleted successfully";
            }
        }

        return "Complaint not found";
    }

}
```

4 ComplaintManagementApplication.java

```java
package com.example.complaintmanagement;

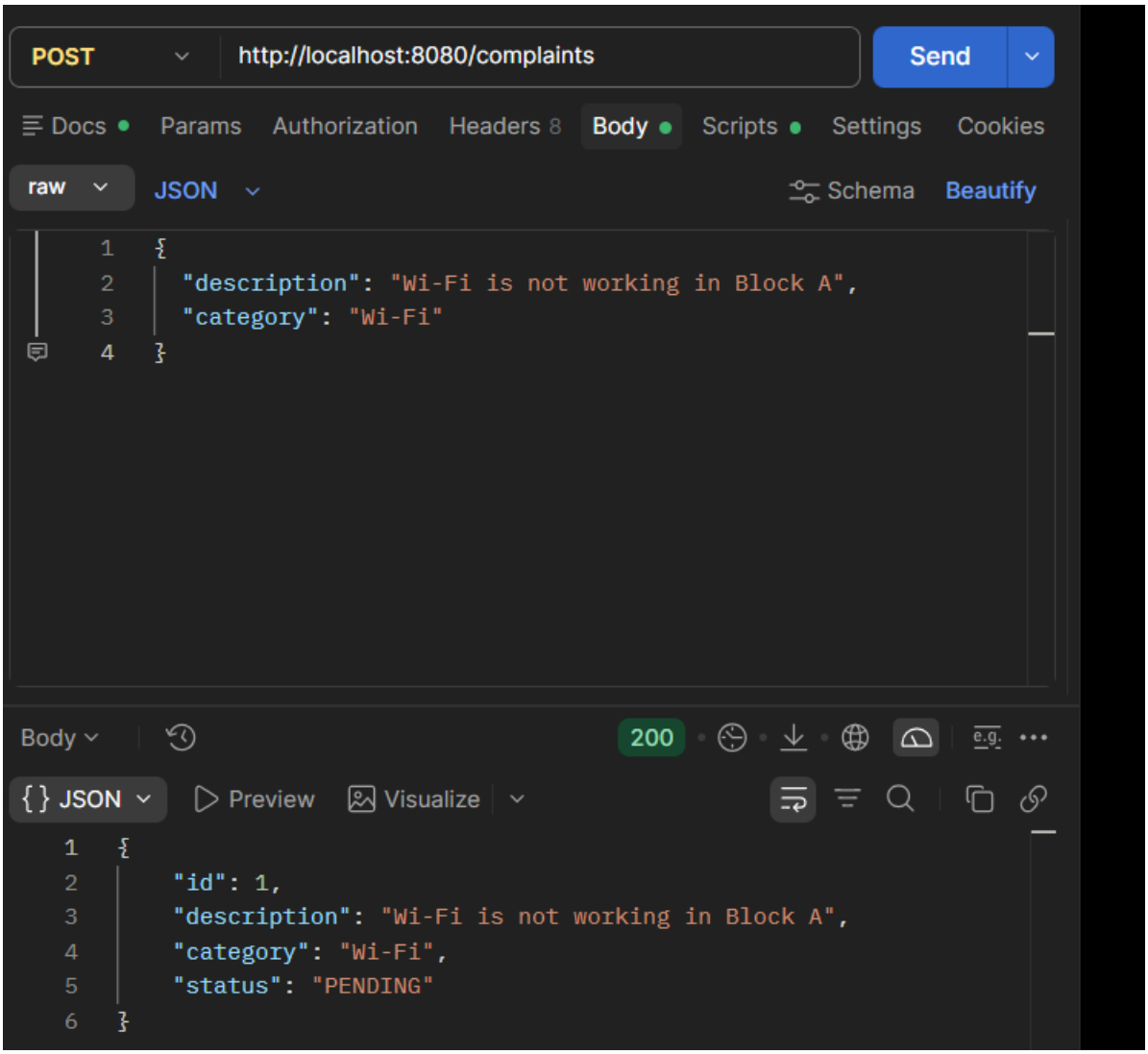
import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class ComplaintmanagementApplication {

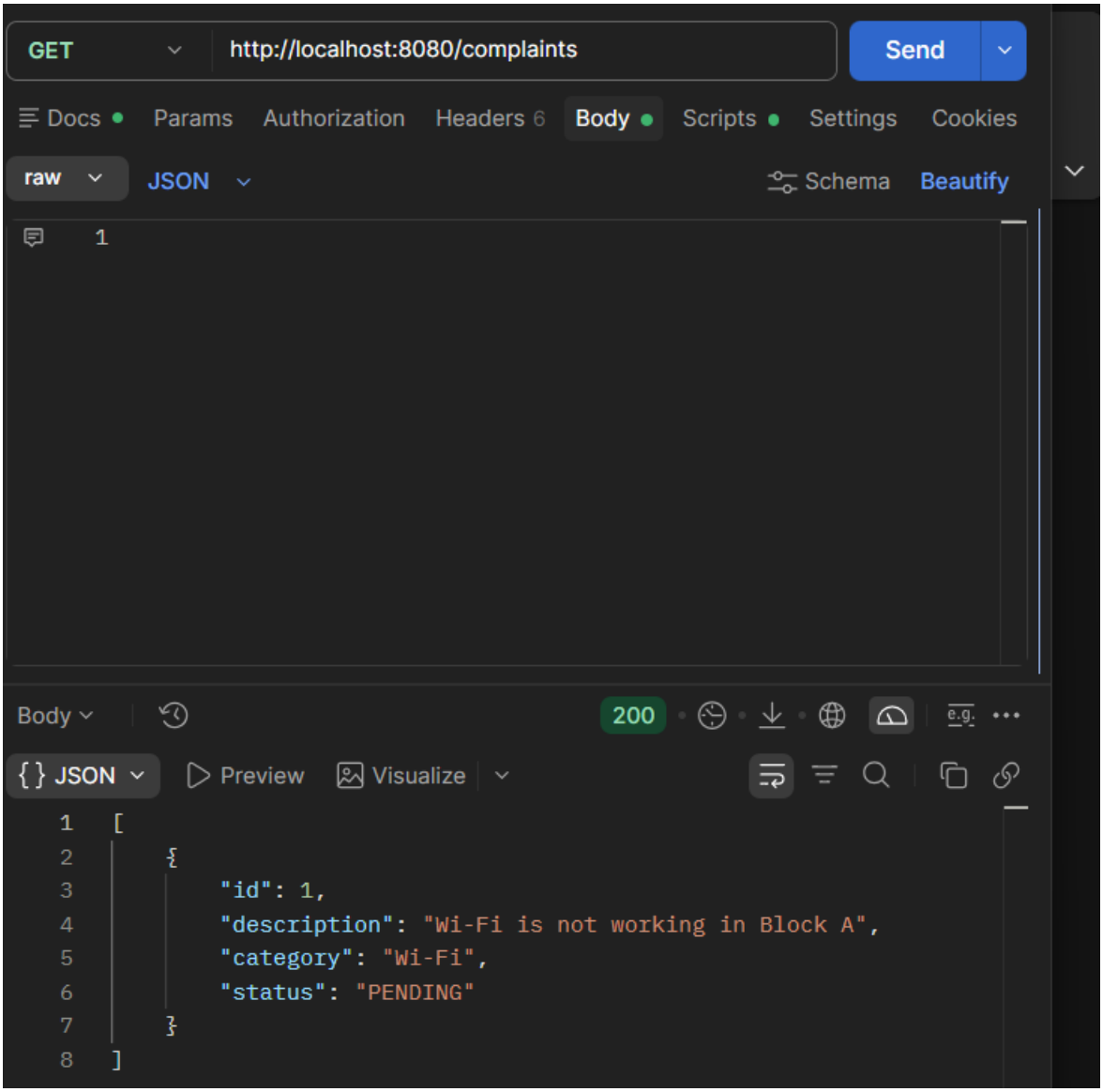
    public static void main(String[] args) {
        SpringApplication.run(ComplaintmanagementApplication.class, args);
    }

}
```

6 post


POST
http://localhost:8080/complaints
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
{
"description": "Wi-Fi is not working in Block A",
"category": "Wi-Fi"
}
Body
200
JSON
Preview
Visualize
{
"id": 1,
"description": "Wi-Fi is not working in Block A",
"category": "Wi-Fi",
"status": "PENDING"
}

7 GET


GET
http://localhost:8080/complaints
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
raw
JSON
Schema
Beautify
1
Body
200
JSON
Preview
Visualize
[
{
"id": 1,
"description": "Wi-Fi is not working in Block A",
"category": "Wi-Fi",
"status": "PENDING"
}
]